SHOAIB

HIL Validation Engineer

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PROFESSIONAL SUMMARY

HIL Validation Engineer with 3+ years of experience developing test plans and executing Hardware-in-the-Loop (HIL) and embedded ECU validation across ADAS, Infotainment, Connectivity, body electronics and Telematics systems. Hands-on with dSPACE HIL benches (AutomationDesk, ControlDesk, ConfigurationDesk), Vector CANoe/CANalyzer, vTESTstudio and VT System, plus CAN/CAN FD/LIN communication, UDS diagnostics and ECU flashing. Builds test automation in Python and CAPL, performs fault injection and trace analysis with CANoe, CANalyzer and ETAS INCA, and reviews Matlab/Simulink control models to derive test conditions. Experienced in requirement-based test design and full traceability from feature specification to test case to result in IBM DOORS, with working knowledge of ISO 26262 and AUTOSAR, within Agile Scrum and ASPICE-driven programs.

TECHNICAL SKILLS

Automotive Domains: ADAS, Infotainment, Connectivity, Body Electronics, Telematics, ECU Validation
Testing: Functional, Integration, System, Regression, Smoke, Fault Injection, Vehicle-Level, HIL, SIL
Standards & Processes: ISO 26262 (Automotive Functional Safety) — working knowledge; AUTOSAR software architecture — working knowledge; Agile Scrum, ASPICE, SDLC, STLC
Protocols: CAN, CAN FD, LIN, UDS, Ethernet
HIL & Test Benches: dSPACE HIL (AutomationDesk, ControlDesk, ConfigurationDesk), Vector VT System, ETAS INCA, VFlash, PCAN, Vehicle Spy
Vector Tools: CANoe, CANalyzer, CANape, vTESTstudio, CAPL
Modeling & Programming: Matlab/Simulink (model review), Python, CAPL, C
Test Automation: Python test scripting, CAPL, vTESTstudio, dSPACE AutomationDesk, requirement-based test design and traceability
Requirement Tools: IBM DOORS, Polarion, PTC Integrity
Defect Tracking: Jira, HP ALM

PROFESSIONAL EXPERIENCE

Chrysler — Auburn Hills, Michigan **Aug 2025 – Present**

HIL Validation Engineer | Project: Infotainment and Connectivity Validation

- Created detailed test plans and validation strategies from feature specifications and system requirements, maintaining requirement-to-test-case-to-result traceability in IBM DOORS.
- Developed and executed test procedures for Bluetooth, USB, Navigation, Audio, Radio, HMI and connectivity features across 8+ vehicle builds, ensuring release readiness.
- Authored and executed 300+ functional, integration, regression and system-level test cases, documenting execution results and defect evidence for engineering review.
- Validated Android Auto and Apple CarPlay across 20+ device/OS combinations.
- Verified Bluetooth profiles (HFP, A2DP, AVRCP, PBAP) and resolved 30+ interoperability defects through structured root-cause analysis.
- Performed navigation route-guidance, destination-search and map validation, plus USB media playback and compatibility testing.
- Executed UDS diagnostic testing and analyzed CAN traces in CANoe and CANalyzer to isolate root causes and track software fixes to closure.
- Developed and maintained Python and CAPL automation scripts and CANoe test configurations to automate regression suites, reducing manual execution effort on each software release.
- Logged, triaged and verified 100+ defects in Jira, driving issues to closure with development and system teams.
- Supported vehicle-level testing and validation of 10+ software releases, coordinating with feature owners to confirm defect elimination.
- Participated in daily Agile Scrum ceremonies, tracked project test metrics, and communicated status, risks and defect trends to feature owners and cross-functional stakeholders.

Environment: *dSPACE, CANoe, CANalyzer, CAPL, Python, CAN, CAN FD, UDS, VFlash, Android Auto, Apple CarPlay, Bluetooth, Jira, DOORS*

Tata Motors — Hyderabad, India **Sep 2021 – Jul 2023**

Automotive Test Engineer | Project: ADAS HIL Validation and Test Automation

- Performed system-level validation of ADAS ECUs against customer requirements in dSPACE HIL environments.
- Developed and executed requirement-based HIL test cases and test plans for ADAS ECU validation, configuring bench setups in dSPACE ConfigurationDesk and running real-time tests through ControlDesk.
- Validated ADAS features including Adaptive Cruise Control, Lane Keep Assist, Automatic Emergency Braking and Blind Spot Detection across simulated sensor inputs and vehicle-state conditions.
- Built automated test sequences in dSPACE AutomationDesk and authored Vector vTESTstudio test cases executed on VT System benches, expanding automated regression coverage across release cycles.
- Performed fault injection testing in dSPACE HIL environments to verify ECU diagnostic and safety responses.

- Analyzed CAN messages and system behavior using CANoe and CANalyzer during HIL validation activities.
- Executed CAN communication testing with Vector tools and verified UDS diagnostic services and fault handling.
- Reviewed Matlab/Simulink control models and feature specifications to derive test conditions and confirm expected ECU behavior during HIL runs.
- Analyzed test failures, performed root cause analysis, and collaborated with development teams to resolve issues.
- Ran regression and integration cycles and generated validation reports and test documentation.
- Participated in requirement reviews in IBM DOORS and maintained traceability from feature specification to test case to test result, supporting defect investigation and resolution.

Environment: *CANoe, CANalyzer, vTESTstudio, VT System, dSPACE (AutomationDesk, ControlDesk, ConfigurationDesk), ETAS INCA, HIL, CAN, UDS, Matlab/Simulink, Python, Jira, DOORS*

EDUCATION

Master of Science, Computer Science — Purdue University **2025**

Bachelor of Technology, Computer Science — Kakatiya Institute of Technology and Science **2022**